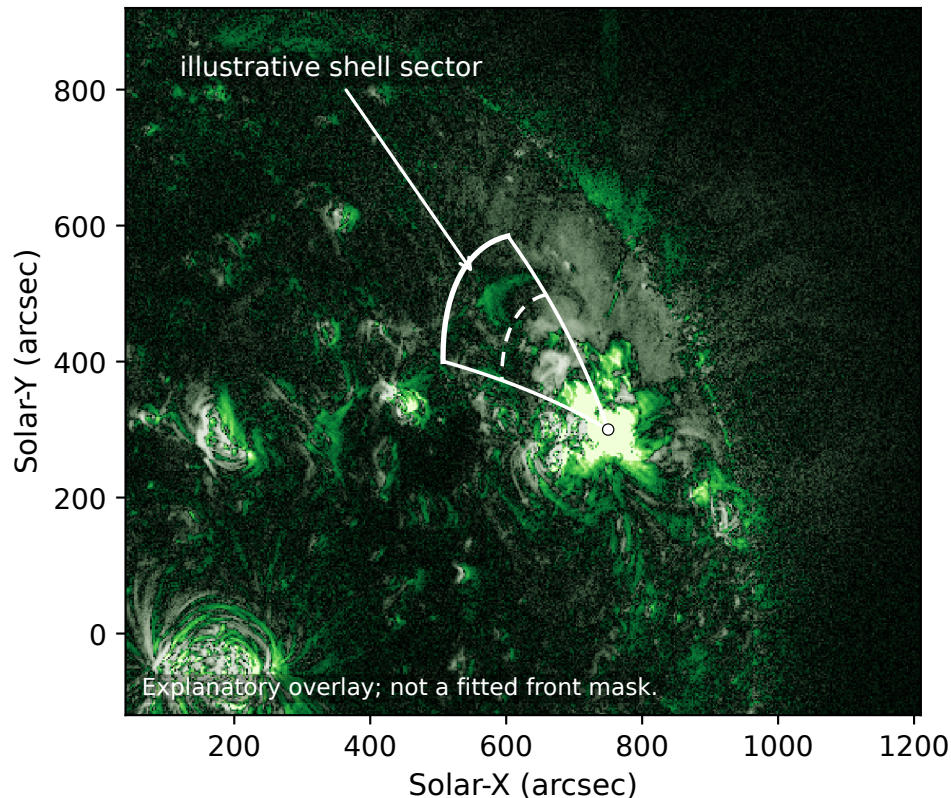
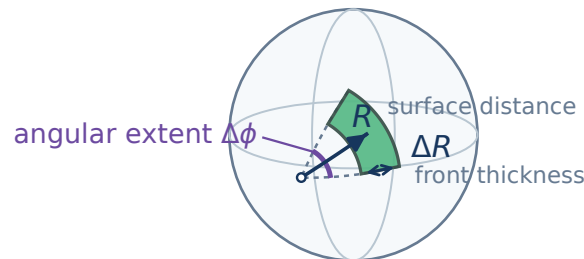


FROM AN OBSERVED FRONT TO AN EXPLICIT VOLUME

(a) Real AIA 193 Å signed base difference



(b) Curved shell sector on the solar surface



$$V \approx R \Delta\phi \Delta R h$$

local surface-volume approximation

half-circumference example: $\Delta\phi = \pi$

front speed

R (Mm)

ΔR (Mm)

h (Mm)

V (cm³)

14 km s⁻¹

40

5

5

3×10^{27}

45 km s⁻¹

80

10

10

2.5×10^{28}



Every energy estimate scales linearly with this assumed volume.